Joint Incident-History Matching for Multilingual Disaster Report Correlation: A Controlled Same-Evidence Study and an Operational Decision Pipeline

Abstract—Disaster reports about the same real-world incident can arrive through different languages and channels, and each report may contain only part of the story. At the same time, separate incidents at one landmark can be described in almost identical words. This study asks whether an incoming report is matched more effectively when the model reads the available history of a candidate incident as one sequence, rather than scoring each earlier report separately and aggregating those scores. We use a Sri Lankan benchmark of 300 incidents and 1,546 reports in Sinhala, Tamil, English, Singlish and Tanglish, across call and SMS styles, with same-landmark hard negatives. A chronological constructor creates 4,085 fixed candidate-incident rows shared by both methods. A jointly fine-tuned mDeBERTa-v3 incident-history matcher is compared with an independently fine-tuned pairwise control using the same historical reports. The joint model achieves higher incident-level AUC on every test split. The largest statistically supported gain appears on the geographically confusable split, from 0.8442 to 0.9638 (Δ = 0.1195; 95% confidence interval 0.0540 to 0.1894). Ablations show that the advantage is concentrated in multi-report histories and same-landmark negatives. The frozen matcher is then used in a learned MERGE/SEPARATE/DEFER policy and tested against a store built by the system itself. Here, the false-merge rate is 0.0224 and cluster purity remains between 0.967 and 0.990, while 61.3% of reports are deferred for review. The complete frozen chain is packaged as a live pipeline and verified against the evaluated system.

Keywords—cost-sensitive decision making, cross-encoder, disaster informatics, event coreference resolution, incident correlation, multilingual natural language processing, online clustering

I. Introduction

Emergency response systems may receive several reports about the same event within minutes and through different channels. In Sri Lanka, for example, one house fire could appear as an English SMS, a Singlish phone call and a Sinhala text message. The reports rarely repeat the same details: one may give the location, another mention casualties, and another describe severity. Some may also be short or noisy. The practical task is to decide whether each new report belongs to an incident already known or describes a separate incident.

That decision is difficult for two main reasons. First, the useful evidence may be scattered across several earlier reports, so a comparison with only one previous report can miss an important connection. Second, separate incidents at the same landmark can use almost the same disaster-related language, which means that lexical or geographic similarity alone may be misleading.

The study therefore asks one controlled research question:

RQ1: Given the same chronologically available evidence for a candidate incident, how does jointly encoding the accumulated incident history compare with pairwise report matching followed by late aggregation, and how does their relative performance change as more evidence becomes available?

The contribution is deliberately focused. Cross-document event coreference, cross-encoders and cluster context are established ideas; this work instead contributes:

- A multilingual disaster benchmark built around difficult cases, including same-landmark negatives, uneven evidence across reports and noisy low-evidence examples.
- A chronological history constructor that uses only reports available before the query, together with a shared evaluation set that gives both methods exactly the same evidence.
- A jointly fine-tuned incident-history matcher evaluated directly against an independently fine-tuned pairwise control.
- Ablation studies that show where the joint model gains an advantage by history size and negative regime, together with a pre-specified calibration extension that was rejected on validation.
- A learned cost-sensitive policy that converts candidate evidence into MERGE, SEPARATE or DEFER decisions without relying on hand-written semantic rules.
- An end-to-end online evaluation in which the system builds and then works from its own incident store, allowing earlier decisions to influence later ones.
- A live implementation whose decision path is checked against the evaluated pipeline across the full benchmark.

Section II reviews related work and identifies the research gap. Sections III and IV describe the benchmark, the common evaluation set and the system architecture. Section V explains the experimental setup, Section VI presents the results, and Section VII covers implementation and verification. Section VIII discusses the findings and limitations, followed by the conclusion in Section IX.

II. Related Work and Research Gap

A. Multilingual and Cross-Document Event Coreference

Earlier work has studied multilingual event sentence coreference using linguistic features [1], while MCECR [6] introduced a multilingual cross-document event coreference dataset. These studies establish the importance of multilingual coreference, but not a streaming setting where the candidate set and available evidence change whenever a new report arrives.

B. Units of Comparison and Cluster Context

Pairwise representation learning has shown that a cross-encoder pair is a strong comparison unit for event coreference [2]. Other studies add document- and topic-level information [3], discourse structure [4], or iterative propagation of inter-event dependencies [5]. In these approaches, broader context supports pairwise scoring or follows it, while the pair remains the main evidence unit.

C. Positioning of This Work

Table I positions this work using the three aspects most relevant to RQ1: multilingual support, the evidence unit being scored, and whether the system consumes its own accumulated state.

TABLE I. POSITIONING AGAINST RELATED WORK

Work	Multi-lingual	Evidence unit	Own state
Tan et al. [1]	yes	sentence pair	no
Yu et al. [2]	no	report pair	no
Xu et al. [3]	no	pair plus context	no
Chen et al. [4]	no	pair plus discourse	no
Choubey and Huang [5]	no	pair, then propagation	no
MCECR [6]	yes	mention pair	no
This work	**yes**	**accumulated history**	**yes**

Two observations matter here. Pairwise comparison remains the main scoring unit, while cluster-level information is usually added as context or through later aggregation. Evaluation is also commonly retrospective over a fixed corpus. A deployed correlation system is different: it can use only evidence that has already arrived and must continue with the state created by its earlier decisions.

This leaves a controlled streaming research gap: with the same chronologically available evidence, how does joint encoding of accumulated incident history compare with pairwise matching and late aggregation, especially as history grows or separate incidents share a landmark?

III. BENCHMARK AND EVALUATION CONSTRUCTION

A. Dataset and Data Contract

The benchmark contains 300 incidents, split evenly between fire and flood, and 1,546 reports with three to eight reports for each incident. It also contains 11,584 supervised report pairs. The reports cover five language forms and two channel styles. Table II summarises the fixed benchmark properties.

TABLE II. FROZEN BENCHMARK PROPERTIES

Property	Value
Incidents	300 (150 fire, 150 flood)
Reports	1,546 (3–8 per incident)
Supervised pairs	11,584, ~30% SAME in every split
Report splits	train 1,076 | val 157 | test_iid 155 | test_geo 98 | test_gen 60
Language forms	Sinhala, Tamil, English, Singlish, Tanglish
Channels	787 SMS, 759 call-style
Hard cases	same-landmark negatives, unequal evidence, low/no-evidence, noisy reports

A data contract determines which fields the models may read and which are reserved for evaluation. Ground-truth incident identifiers are structurally blocked from the model and decision path. Negatives include same-landmark same-subtype pairs, same-subtype pairs from different locations, and cross-type pairs, reducing shortcuts based on location or incident type. Arrival-time distributions were also audited so that time could not act as an unintended label. A time-only gradient-boosted model achieves 0.5075 AUC on test_iid, effectively chance performance.

B. Chronological History Assembly

For every query report and candidate incident, the history contains only reports that arrived before the query. Earlier reports are kept in chronological order and limited by a 500-token whole-report recency budget. If the limit is exceeded, the oldest reports are removed first. The same history-building rule is used during both evaluation and deployment so that the evidence presented to the model remains consistent across stages.

Both methods use the same fixed set of (query report, candidate incident, SAME/DIFFERENT) rows. During manual auditing, we found that the constructor could draw the same candidate incident more than once for a query. This produced 211 duplicate rows across 183 groups: the content matched, but different row identifiers unintentionally changed the weighting. We fixed the constructor to exclude already selected candidates, rebuilt the affected stages, and froze the corrected evaluation set at 4,085 rows with 1,246 positives and no duplicate units before comparison.

IV. SYSTEM ARCHITECTURE

Fig. 1 shows the closed-loop process followed for each incoming report, and Table III lists the same stages in order. The system maintains its own incident store, so each decision changes the evidence that will be available for the next report. This feedback loop is what makes the setting different from independent, one-off classification.

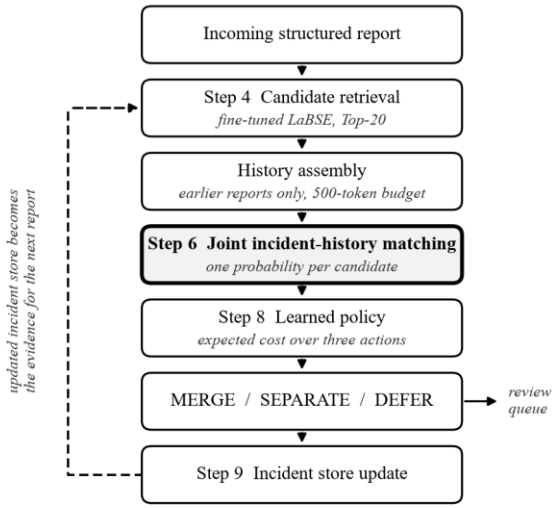


Fig. 1. Closed-loop live inference. The shaded stage is the research contribution; the dashed path is what makes the system dynamic, because the store a decision produces is the evidence the next report is scored against.

TABLE III. LIVE INFERENCE STAGES

Stage	Input	Output
Retrieval	New report and current store	Top-20 candidate clusters with similarities
History assembly	Each candidate cluster	Chronological earlier-report history under the token budget

Stage	Input	Output
Joint matching	Query and full candidate history	One membership probability per candidate
Policy	Candidate scores, query-level features, urgency	MERGE to a named cluster, SEPARATE, or DEFER
Store update	Action and incoming report	Updated store and audit trace for the next report

A. Candidate Retrieval

Candidate retrieval uses a LaBSE bi-encoder fine-tuned with multiple-negatives ranking loss and hard negatives (learning rate 1e-5, one epoch, batch size 32). The checkpoint is chosen using validation Recall@5. For each incident, the query is compared with the reports currently stored in that incident, and the highest cosine similarity becomes the incident score. The 20 highest-scoring incidents are passed to the next stage. Retrieval does not decide whether two reports belong together; its purpose is to keep the correct incident in the candidate set whenever possible.

B. Pairwise Control

The pairwise control uses an independently fine-tuned mDeBERTa-v3 cross-encoder trained on 8,020 report pairs. A predeclared sweep evaluates learning rates of 1e-5, 2e-5 and 3e-5 for three epochs each. During inference, the query is scored separately against every earlier report in a candidate history, after which the probabilities are aggregated. MAX was specified in advance as the main aggregation rule. Scoring the frozen evaluation set requires 11,115 cross-encoder calls. This makes the pairwise method a strong conventional control rather than a deliberately weak baseline.

C. Joint Incident-History Matching

The proposed method changes what the cross-encoder sees as one example. Instead of scoring a query against earlier reports one at a time, it receives the full chronological history of a candidate incident and returns one probability that the query belongs there. Training uses 2,863 naturally constructed joint rows, with no pairwise augmentation, and a maximum sequence length of 512. All 4,085 frozen evaluation inputs fit within this limit; the longest has 503 tokens. Model selection considers three learning rates and three checkpoints per rate using validation incident-level AUC only. The joint formulation therefore needs 4,085 forward passes, compared with 11,115 calls for the pairwise control.

D. Learned Operational Policy

A separate decision layer turns candidate scores into an operational action. The decision is made for the report together with its full candidate set, using the highest score, second-highest score, their margin, and the number of candidates. History size was also audited on training data, but because it carried label information at 0.1757 AUC, it was removed rather than used as a shortcut. Two probability heads estimate whether the top candidate is correct and whether the report belongs to any existing incident. These probabilities become expected costs for MERGE, SEPARATE and DEFER. False-merge cost is scaled by urgency (2, 5 and 10 for low, medium and high) as an operational safety preference, not a learned value. Urgency is never used by the matcher as an identity feature. A 16-hidden-unit multilayer perceptron was selected over logistic regression on validation data.

Let P_top and P_exist denote the outputs of the two probability heads, and let u denote the urgency of the incoming report. The expected cost of each action is

$$C_{merge} = (1 - P_{top}) \cdot C_{fm}(u), \quad C_{sep} = P_{exist} \cdot C_{fs}, \quad C_{defer} = C_d(u), \tag{1}$$

where C_fm(u) is the urgency-dependent cost of a false merge, C_fs is the cost of a false separation, and C_d(u) is the cost of deferring. The selected action is

$$a^* = \arg\min C_a, \tag{2}$$

The selected action is the one with the lowest expected cost. Because C_fm(u) increases with urgency while C_fs stays fixed, identical model probabilities can lead to a more cautious decision for a high-urgency report than for a low-urgency one.

E. Online State Update

The chosen action is then applied to the incident store. MERGE places the report in exactly one existing cluster, SEPARATE creates a new cluster, and DEFER sends the report to a review queue without adding it to the automatic store. When no candidate exists, the report is treated as SEPARATE because there is nothing available to merge with. This is a deterministic cold-start rule, not a model prediction.

V. Experimental Setup

A. Corpus and Splits

All experiments use the benchmark in Table II: 300 incidents, 1,546 reports and 11,584 supervised pairs. The report split contains 1,076 training, 157 validation, 155 in-distribution test, 98 geographic test and 60 generalisation test reports, with about 30% positive pairs in each split. The controlled comparison in Section VI-B uses the 4,085 frozen candidate-incident rows from Section III-B, including 1,246 positives. The benchmark is constructed rather than collected from live emergency traffic, as discussed in Section VIII.

B. Baselines and Metrics

Three comparators are used. The fine-tuned bi-encoder provides the retrieval ceiling, the independently fine-tuned pairwise cross-encoder is the same-evidence control, and a fixed-threshold rule is the baseline for the learned policy. Matching is measured with incident-level AUC, while retrieval uses Recall@1, Recall@5 and Recall@20. The operational policy is evaluated with false-merge, false-separate and defer rates, coverage and mean expected cost. The final incident store is evaluated using CoNLL F1 and weighted cluster purity.

C. Implementation

Both cross-encoders are fine-tuned from mDeBERTa-v3-base, while the bi-encoder is fine-tuned from LaBSE. The implementation uses PyTorch 2.13, Transformers 5.14, sentence-transformers 5.6 and Python 3.11. Fine-tuning was performed on a single NVIDIA T4 GPU. Evaluation, the operational policy and the live pipeline run on CPU in fp32. We do not use mixed precision because fp16 changed a joint score by 2.5×10^{-2}, which is large compared with the score differences being analysed. Dependencies are pinned and model checkpoints are frozen; Section VII describes the runtime checks used to verify them.

D. Protocol and Statistical Testing

The same protocol was followed at every stage. Each procedure was specified before execution, model and hyperparameter selection used validation data only, and every test set was scored once. No stage was retrained because of a disappointing test result. Data splits remain isolated, and evaluation-only fields are structurally unavailable to the models. The primary comparison uses a predeclared, time-matched same-landmark subset to reduce arrival-time and place-name shortcuts. Confidence intervals use a paired bootstrap clustered by query because both methods score the same rows and each query contributes multiple candidates.

VI. Results

A. Candidate Retrieval

For live-like test queries that are not cold starts, retrieval reaches Recall@1 = 0.7787, Recall@5 = 0.9407 and Recall@20 = 0.9881. At the top-20 stage, the correct incident is retained for 250 of 253 queries. Only three queries lose the correct incident before joint scoring, so retrieval failure contributes little to the measured end-to-end result. That sample is also too small to support a separate conclusion about the behaviour of retrieval errors.

B. Controlled Head-to-Head Comparison

Table IV gives the main controlled comparison. Both models receive exactly the same candidate histories; the difference lies only in how they process that evidence.

TABLE IV. Incident-level AUC on the primary time-matched, same-landmark subset

Split	n	LaBSE	Pairwise	Joint	Δ
val	134	0.8233	0.9454	—	—
test_iid	116	0.8056	0.9013	0.9358	+0.0345
test_geo	86	0.7485	0.8442	0.9638	+0.1195
test_gen	50	0.9024	0.9712	0.9968	+0.0256

The joint model has a higher AUC point estimate on every test split. The largest improvement appears on the geographically confusable split, where the 95% confidence interval for the gain is [+0.0540, +0.1894]. The in-distribution split also shows a positive estimated gain, but its interval [-0.0092, +0.0872] crosses zero, so that result is not statistically conclusive. The generalisation split is reported as a secondary result.

Two limits matter. Both models receive the same historical reports, so the joint model has no extra evidence. However, they are independently fine-tuned using different training formulations. The observed difference therefore cannot be attributed to evidence accumulation alone.

C. Where the Advantage Arises

Table V groups the same frozen test scores by the number of earlier reports available in each candidate incident history, pooling the test splits as specified in advance.

TABLE V. AUC by history size (pooled test)

History size	n	Joint	Pairwise	Δ	95% CI
1	218	0.9417	0.9330	+0.0086	[-0.0153, +0.0340]
2	191	0.9889	0.9752	+0.0137	[+0.0006, +0.0307]
3–4	299	0.9950	0.9724	+0.0227	[+0.0094, +0.0404]
5+	101	0.9935	0.9810	+0.0126	[+0.0005, +0.0382]

With only one earlier report, the difference between the two methods is indistinguishable from zero, which is reasonable because there is no additional history to combine. Once two or more earlier reports are available, the joint model shows a positive, interval-supported gain in every history-size group. The pattern is consistent with a benefit from joint history, although it does not identify a single causal mechanism.

Table VI groups the same frozen scores by the relationship between the query and each negative candidate.

TABLE VI. AUC by negative regime (pooled test)

Negative regime	n	Joint	Pairwise	Δ	95% CI
same landmark	425	0.9574	0.8982	+0.0592	[+0.0378, +0.0833]
same subtype	467	0.9967	0.9841	+0.0126	[+0.0048, +0.0220]
cross type	423	0.9975	0.9871	+0.0103	[+0.0037, +0.0186]

The improvement on same-landmark negatives is about five times larger than the improvement on cross-type negatives. Read together with Table V, this places the strongest advantage in the hardest operating condition: several reports already describe one incident while a different nearby incident shares the same location name and much of the same vocabulary.

We also evaluated an optional calibrator that added a capped arrival-time gap to the joint score. Both the extension and its acceptance rule were specified before evaluation. It raised the overall validation score from 0.9903 to 0.9909, but reduced performance on the time-matched same-landmark stratum from 0.9822 to 0.9813. Because it failed the validation gate, the calibrator was rejected and was not tested on the test sets.

D. Operational Policy

The learned policy was evaluated on the actual retrieved candidate sets for all 313 test reports. These include 60 observed cold-start reports (19.2%), which provide genuine new-incident cases within the benchmark, as well as retrieval misses. Table VII compares the learned policy with a naive fixed-threshold decision rule.

TABLE VII. Policy operating point on pooled test (313 reports)

Metric	Learned policy	Naive argmax at 0.5
False-merge rate	0.0319	0.1438
False-separate rate	0.0735	—
DEFER rate	0.3131	0.0000
Coverage	0.6869	1.0000
Mean cost	0.3604	0.6997

Compared with the naive rule, the learned policy cuts the false-merge rate by roughly a factor of four and nearly halves the mean expected cost, while deferring about one third of reports. The trade-off behaves as expected: assigning a higher cost to false merges lowers the false-merge rate but increases deferral. Cold starts remain a notable risk. Their false-merge

rate is 0.1167, representing seven genuinely new test incidents that were incorrectly absorbed into existing candidates.

E. End-to-End Online Evaluation

The full frozen pipeline was then replayed chronologically against a store built from the system's own earlier decisions, so every decision changes the evidence available later. Across all 1,546 reports, 582 were placed automatically and 964 were deferred, an automatic placement rate of 37.6%. Table VIII compares the pooled test operating point under gold-built histories with the operating point under the system-created store.

TABLE VIII. GOLD-BUILT HISTORIES VERSUS SYSTEM-CREATED STORE (313 TEST REPORTS)

Metric	Gold histories	System store
False-merge rate	0.0319	0.0224
False-separate rate	0.0735	0.0990
DEFER rate	0.3131	0.6134
Coverage	0.6869	0.3866

The two columns in Table VIII describe different operating conditions. The first uses gold-built incident histories, while the second reflects the state the system actually produced. When deferred reports are kept as unresolved singletons, CoNLL F1 is 0.2789, 0.2531 and 0.3066 across the three test splits. If only automatically placed reports are considered, the corresponding scores are 0.6897, 0.6417 and 0.8139. Weighted cluster purity remains between 0.967 and 0.990.

Table IX shows how the measured decision signals change under the system-created store and helps explain the drop in coverage.

TABLE IX. DECISION SIGNALS UNDER GOLD HISTORIES AND SYSTEM STATE

Signal	Gold	System	Change
Mean top score	0.8290	0.8535	+0.0245
Mean runner-up score	0.2969	0.6393	+0.3424
Mean margin	0.5321	0.2142	−0.3179
Mean P(top correct)	0.6878	0.5177	−0.1701
Runner-up above 0.9	20.8%	48.9%	—

The main difference is not a weaker top match but a much stronger runner-up. If one real incident is split across several system-created clusters, multiple candidates can score highly at once. The margin shrinks, which the frozen policy reads as ambiguity and therefore defers more often. In-loop target-cluster Recall@1 falls to 0.4804, 0.3676 and 0.6857 across the three test splits, compared with 0.7787 under gold-incident retrieval. Of 38 test errors, 21 occur when the system is already working from a damaged state. The matcher is not simply becoming weaker; fragmentation changes the ranking environment in which the policy must decide.

VII. IMPLEMENTATION AND VERIFICATION

The frozen chain is implemented as a live pipeline. A structured report enters through an input adapter, the four processing stages run against the current incident store, and a browser dashboard shows the candidate set, final decision, expected costs, incident store, review queue and per-report audit trace. The same trace is printed to the console as each stage finishes.

Two safeguards keep deployment aligned with evaluation. First, the input adapter accepts only the seven fields in the data contract and rejects extras, preventing ground-truth identifiers from entering the operational path. Report text is validated on a stripped copy but stored unchanged, so preprocessing remains consistent. Second, the live pipeline imports the store logic, history constructor and policy forward pass from the evaluated implementation rather than re-implementing them, reducing the risk of behavioural drift.

Incoming reports do not contain precomputed embeddings in deployment, so the frozen bi-encoder generates them online. Before the pipeline is allowed to make decisions, three identity checks are performed. The policy forward pass must reproduce 47 stored reference vectors within 1e-9; online embeddings for 128 benchmark reports must match the frozen vectors with cosine similarity above 0.999; and the joint matcher must reproduce 128 stored scores within 1e-3. Table X reports the observed values for both GPU and CPU runtimes.

TABLE X. IDENTITY GATES ON TWO RUNTIMES

Gate	Tolerance	GPU	CPU
Policy self-test	1e-9	0.00e+00	0.00e+00
Embedding identity	cos > 0.999	1.000000	1.000000
Matcher identity	1e-3	3.97e-06	1.53e-05

Decision equivalence is also checked directly. All 1,546 benchmark reports are replayed twice with the same stub scorer: once through the evaluation replay and once through the deployed engine. The two runs must agree on 20 fields for every report, as well as on the final clusters and the set of deferred reports. Console tracing only observes values that have already been computed, and the equivalence test is also run without tracing, so the instrumentation cannot change a decision. On commodity CPU hardware, end-to-end latency for the demonstration store ranges from 0.2 to 0.7 seconds per report.

VIII. DISCUSSION

The controlled comparison supports joint incident-history encoding when both methods receive the same evidence, with the clearest result on geographically confusable incidents. The ablations also place the improvement where it is most useful: when several earlier reports are available and when negative candidates share a landmark. The findings still require care. The in-distribution gain is not statistically conclusive, and the models differ in training formulation as well as evidence unit.

The online evaluation shows something static gold-history experiments cannot: the system becomes more cautious when it must use the store created by its own earlier decisions. Fragmenting one incident across several clusters creates multiple plausible candidates and narrows the gap between the top two scores, so the policy defers more cases. Deferred reports are not discarded; they enter a review queue with their candidate clusters, scores and probabilities for human review.

This trade-off is appropriate in disaster response because the two error types have different consequences. Merging separate emergencies may hide the need for a distinct response, while a deferred report can still be escalated to an operator. The cost structure represents this asymmetry directly. Low false-merge rate and high cluster purity, with lower automatic coverage, are therefore an intentional safety trade-off.

A. Limitations

Four limitations define the scope of these findings. First, the in-distribution improvement is not statistically conclusive, so the strongest head-to-head evidence comes from the geographic split. Second, the models differ in training formulation as well as evidence unit, so the gain cannot be attributed only to evidence accumulation. Third, the benchmark is constructed rather than collected from live emergency traffic, avoiding privacy risks but leaving real-world transfer untested. Finally, only three test queries lose the correct incident during retrieval, too few for reliable conclusions about retrieval-failure behaviour.

Reproducibility is supported by frozen checkpoints, pinned dependencies and runtime identity checks. Each report also records an audit trace covering retrieval, joint scoring, policy features, expected costs and the resulting store update, while a shared history constructor reduces maintenance drift. Because the benchmark uses constructed reports rather than real emergency communications, no personal data from affected individuals was processed. In deployment, uncertain cases should remain subject to human review.

IX. Conclusion

This study asked whether multilingual disaster reports are better correlated by jointly encoding the full chronological history of a candidate incident or by reducing that history to separate pairwise comparisons. With the same evidence, the joint model achieves higher AUC on every test split and a statistically supported improvement from 0.8442 to 0.9638 on the geographically confusable split. Ablations place the advantage in multi-report histories and same-landmark negatives, where evidence can be combined and simple lexical similarity is least reliable.

A second finding appears when the frozen matcher is extended with a learned cost-sensitive decision policy and evaluated chronologically against a system-created store. As earlier mistakes fragment the store, the system becomes more cautious. The false-merge rate remains low at 0.0224 and weighted cluster purity stays between 0.967 and 0.990, but many ambiguous reports are deferred for human review. The measured reason is the growing strength of competing runner-up candidates rather than a simple decline in the top matcher score. The resulting pipeline runs live, and its decisions are verified to match the evaluated implementation across the complete benchmark.

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